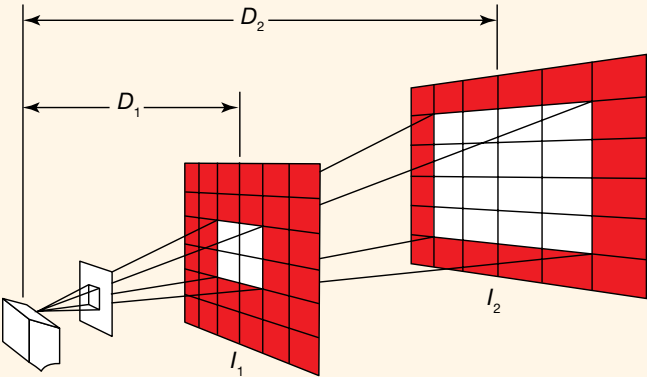


Inverse Square Law

(Eq. 1) $I_1 = \frac{D_2^2}{D_1^2}$

where
 I_1 is the intensity at D_1 ,
 I_2 is the intensity at D_2 ,
 D is the distance from source.



Milliamperage–Time

(Eq. 1) $\frac{M_1}{M_2} = \frac{T_2}{T_1}$

or

(Eq. 2) $M_1 \times T_1 = M_2 \times T_2$

where
 M is the milliamperage,
 T is the exposure time.

Activity–Distance

(Eq. 1) $\frac{A_1}{A_2} = \frac{D_1^2}{D_2^2}$

where
 A is the activity of isotope
in curies (becquerels),
 D is the distance from
source.

To solve for new intensity:

(Eq. 2) $I_2 = \frac{I_1 \times D_1^2}{D_2^2}$

To solve for new distance:

(Eq. 3) $D_2 = \sqrt{\frac{I_1 \times D_1^2}{I_2}}$

Milliamperage–Distance

(Eq. 1) $\frac{M_1}{M_2} = \frac{D_1^2}{D_2^2}$

where
 M is the milliamperage,
 D is the distance from source.

Time versus Curies

To calculate a new exposure
time (T_2) when the number of
curies (Ci) changes:

(Eq. 1) $T_2 = \frac{Ci_1 \times T_1}{Ci_2}$

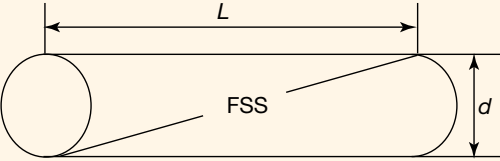
Time versus Source-to-Film
Distance

To calculate new exposure time (T_2)
when SFD changes:

(Eq. 1) $T_2 = \frac{T_1 \times (SFD_2)^2}{(SFD_1)^2}$

Focal Spot Size

(Eq. 1) $FSS = \sqrt{L^2 + d^2}$
where
 d is the source diameter,
 L is the source length.



Time versus Density

(Eq. 1) $T_2 = \frac{D_2 \times T_1}{D_1}$

where
 T_2 is the exposure time to
achieve desired density D_2 ,
 T_1 is the original exposure
time,
 D_1 is the unacceptable
density (too light or
too dark).

Time–Distance

(Eq. 1) $\frac{T_1}{T_2} = \frac{D_1^2}{D_2^2}$

where
 T is the exposure time,
 D is the distance from
source.

Source-to-Film Distance

Traditional unit: $U_g = 0.02$ in.

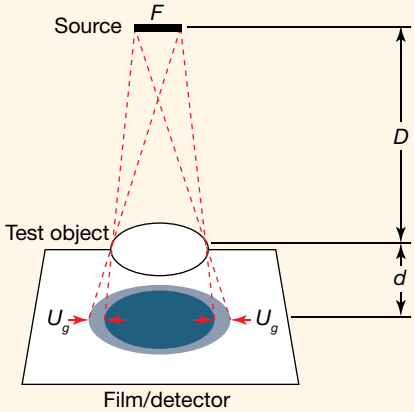
(Eq. 1) $SFD = \frac{F \times d}{0.02} + d$

SI unit: $U_g = 0.5$ mm

(Eq. 2) $SFD = \frac{F \times d}{0.5} + d$

where
 F is the source or focal spot
size,
 d is the source side of
object-to-film distance
(object thickness).

Geometric Unsharpness



(Eq. 1) $U_g = \frac{F \times d}{D}$

where
 U_g is the geometric unsharpness,
 F is the source or focal spot size,
 d is the source side of object-to-
film distance (object thickness),
 D is the source-to-object distance
(SOD).

Quality Factors of Radiation

Radiation	Quality Factor
X-ray	1
Gamma ray	1
Beta particles	1
Thermal neutrons	5
Fast neutrons	10
Alpha particles	20

RT

REFERENCE CARD

Catalog Number: 10141

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Equivalent Sensitivity

Per ASME BPVC, Section V, Article 22, equivalent IQI sensitivity is that thickness of the IQI, expressed as a percentage of the part thickness (T_m), in which the 2T hole would be visible under the same conditions.

Equivalent sensitivity	Quality level	IQI (T as percent of T_m)	Perceptible hole
0.7%	1-1T	1%	1T
1%	1-2T	1%	2T
1.4%	2-1T	2%	1T
2%	2-2T	2%	2T
2.8%	2-4T	2%	4T
4%	4-2T	4%	2T

Radiographic Conversions

Radiation	Traditional Units	SI Units	Conversion
Exposure	roentgen (R)*	coulombs per kg (C/kg)	1 C/kg = 3876 R 1 R = 2.58 × 10 ⁻⁴ C/kg
Activity	curie (Ci) 1 Ci = 2.22 × 10 ¹² dpm	becquerel (Bq) 1 Bq = 1 dps	1 Ci = 3.7 × 10 ¹⁰ Bq 1 Bq = 2.7 × 10 ⁻¹¹ Ci
Absorbed Dose	rad 1 rad = 100 ergs/g	gray (Gy) 1 Gy = 1 J/kg	1 rad = 0.01 Gy 1 Gy = 100 rad
Dose Equivalent	rem rem = rad × QF	sievert (Sv) Sv = Gy × QF	1 rem = 0.01 Sv 1 Sv = 100 rem

*For practical radiography using gamma and X-radiation, 1 R is considered equivalent to 1 rem and is also converted to sieverts.

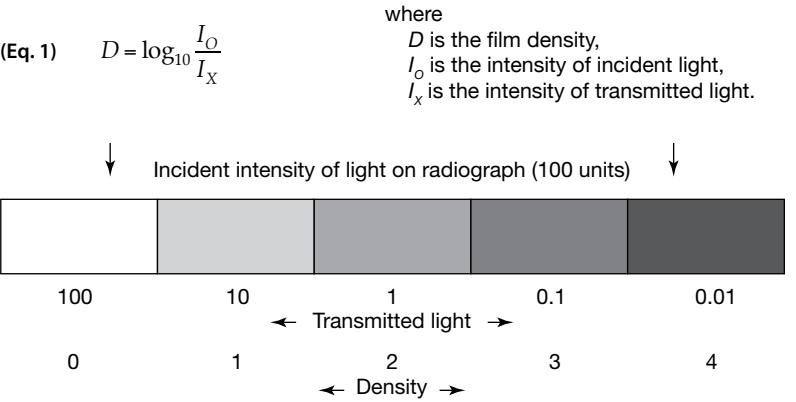
Gamma-Ray Sources

Isotope	Half-Life	Photon Energy (MeV)	Radiation Output –R/h/Ci@ 1 ft (RHM/Ci)	Focal Spot Size – in. (mm)	Practical Source (Curies)
Ir-192	74 d	0.21–0.61 (12 rays)	5.2 (0.48)	0.3 (7)	200
Co-60	5.3 y	1.17 and 1.33	14.0 (1.3)	0.5 (12)	100
Tm-170	128 d	0.054 and 0.084	0.033 (0.0025)	0.1 (3)	50
Cs-237	30.2 y	0.66	3.4 (0.32)	0.4 (10)	75
Se-75	120 d	0.066 and 0.612	14.0 (0.203)	0.3–1.2 (8–30)	75

Penetrating Power in Steel

Source	Effective Energy Level	Thickness of Steel – in. (mm)
Selenium 75	0.22 MeV	Up to 1.25 (31.75)
Iridium 192	0.31–0.47 MeV	0.25–3.5 (6.35–89)
Cobalt 60	1.17 and 1.33 MeV	3–7.75 (76–198)
X-rays	150 kV	0.75–1.25 (19–31.75)
	250 kV	1.5–2.25 (38–57)
	400 kV	2.75–4 (70–101.6)
	1 MeV	3.5–8 (89–203)

Radiographic Film Density



Properties of Digital Radiographic Detectors

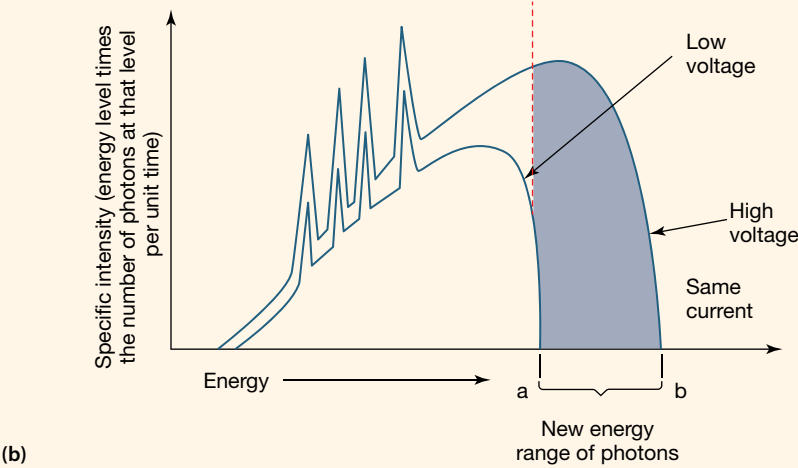
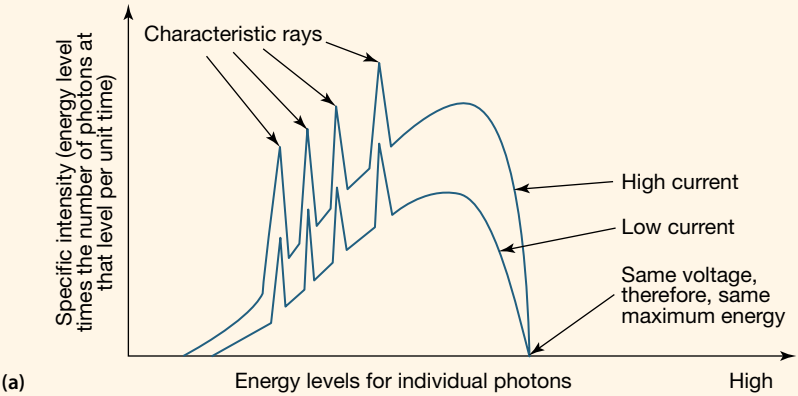
Detector	Size Range (mm)	Pixels	Square Pixel Size (µm)	Image Acquisition Speed (min)
Amorphous silicon	200 × 200 230 × 190 410 × 410 280 × 410	1024 × 1024 2304 × 1920 2048 × 2048 2304 × 3200	100 127 200	< 1 (real time)
Amorphous selenium	350 × 430	≤ 2560 × 3072	139	< 1.0–1.5
Charge-coupled devices	≤ 100 × 100 larger with lenses	≤ 4096 × 4096	9–50 larger with optics	< 1 (real time)
Storage phosphors	≤ 1550 × 430	≤ 15 500 × 4300	25–250	< 1–4
Linear arrays	≤ 500	≤ 4096	10–200	< 1 (real time)
Reversed geometry	≤ 450 × 450	≤ 2048 × 2048	25–200	< 1 (real time)

Personnel Dosimetry Devices

Device	Uses	Measures	Units
Film Badge	Whole-body dosimetry	Beta (high E) Gamma/X-rays*	Dose equivalent (millirem)
TLD/OSL	Whole-body and ring badges	Gamma/X-rays	Dose equivalent (millirem)
Pocket Dosimeter	Elevated radiation areas; immediate results	Gamma/X-rays	Exposure (mR)
Digital Dosimeter	Elevated radiation areas; immediate results; audible warning	Gamma/X-rays	Exposure (mR)
Can be made to measure neutrons			

Effects of Kilovoltage and Amperage

Kilovoltage	Low amperage	High amperage
Low kilovoltage	Low-intensity soft X-rays	High-intensity soft X-rays
High kilovoltage	Low-intensity hard X-rays	High-intensity hard X-rays



Effect of a change (a) in current on X-ray spectrum; (b) in voltage on X-ray spectrum.

Occupational Dose Limits

Organ/Region	NRC Limit (per year) – rem (mSv)
Whole Body	5 (50)
Lens of Eye	15 (150)
Extremities	50 (500)
Skin	50 (500)
Embryo/Fetus	0.5 (5)
General Public	0.1 (1)

Degree of Enlargement

(Eq. 1) $\frac{S_o}{S_i} = \frac{D_o}{D_i}$

where S_o is the object size, S_i is the image (shadow) size, D_o is the source-to-object distance (SOD), D_i is the source-to-film distance (SFD).